

seconds hand

Harrison's H1 chronometer

John Harrison's first "sea clock" was the H1, which he made to solve the longitude problem—how to measure time accurately enough at sea to calculate longitude. However, the H1 was impractically large, a problem Harrison solved in 1759 with his H4 chronometer.

minute hand

1735–59

The chronometer

John Harrison makes the first marine chronometer (the H1) in 1735. He then makes improved versions, culminating in the H4 in 1759.

calendar hand,
indicating date
of the month

hour hand

1907

Gyroscopic compass

American Elmer Sperry invents the gyroscopic compass, a major advance for accurate navigation because it always points to true north and is not subject to deviation.



Ship's compass

1930s–40s

Radar

The invention of radar makes it possible to determine an object's position even when it cannot be seen.



Radarscope

Late 20th century

Global positioning systems

The introduction of satellite-based GPS makes it possible to pinpoint locations and navigate to within a few yards.



GPS chart plotter